

Warm-Up Quiz — Day 13

Moment Conditions, Rank, and the Method of Moments

Complete before lecture. 10 minutes.

Name: _____

Today we introduce instrumental variables. The IV estimator is derived from moment conditions $\mathbb{E}[\mathbf{z}(Y - \mathbf{x}'\beta)] = 0$, and identification requires a rank condition on $\mathbb{E}[\mathbf{z}\mathbf{x}']$. This quiz reviews the algebra of moment conditions and matrix rank.

1. OLS as Method of Moments.

The OLS estimator solves $\frac{1}{n}\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$, i.e., $\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i(y_i - \mathbf{x}_i'\hat{\beta}) = 0$.

- (a) This is the sample analog of which population condition?
- (b) Solve the sample condition for $\hat{\beta}$. (You should get the OLS formula.)

2. When OLS Fails: Endogeneity.

Suppose $Y_i = \beta X_i + e_i$ but $\text{Cov}(X, e) = \sigma_{Xe} \neq 0$.

- (a) Show that $\text{plim } \hat{\beta}_{OLS} = \beta + \sigma_{Xe}/\text{Var}(X)$. (Hint: $\hat{\beta} = \beta + \frac{\frac{1}{n}\sum X_i e_i}{\frac{1}{n}\sum X_i^2}$.)
- (b) If higher income causes both more education (X) and better health (Y), what is the sign of σ_{Xe} ? Is $\hat{\beta}$ biased up or down?

3. Switching the Instrument.

Now suppose we have an instrument Z with $\text{Cov}(Z, e) = 0$ and $\text{Cov}(Z, X) \neq 0$. The IV moment condition is $\mathbb{E}[Z(Y - \beta X)] = 0$.

- (a) Solve this population moment condition for β . (Express β in terms of covariances.)

- (b) Write the sample analog and solve for $\hat{\beta}_{IV}$.

4. Matrix Rank (Preview for Identification).

The IV identification condition requires $\text{rank } \mathbb{E}[\mathbf{z}\mathbf{x}'] = k$.

- (a) $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$. What is $\text{rank}(\mathbf{A})$? Is $\mathbf{A}$ invertible?

- (b) $\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. What is $\text{rank}(\mathbf{B})$? Is $\mathbf{B}$ invertible?

Answer Key — Day 13

1. (a) $\mathbb{E}[\mathbf{x}(Y - \mathbf{x}'\beta)] = \mathbf{0}$, i.e., $\mathbb{E}[\mathbf{x}e] = \mathbf{0}$.
 (b) $\frac{1}{n}\mathbf{X}'\mathbf{y} = \frac{1}{n}\mathbf{X}'\mathbf{X}\hat{\beta} \implies \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.
2. (a) $\text{plim } \hat{\beta} = \beta + \frac{\text{plim } \frac{1}{n} \sum X_i e_i}{\text{plim } \frac{1}{n} \sum X_i^2} = \beta + \frac{\mathbb{E}[Xe]}{\mathbb{E}[X^2]} = \beta + \frac{\sigma_{Xe}}{\text{Var}(X)}$ (using $\mathbb{E}[X] = 0$ wlog).
 (b) Income is positively correlated with both X and e , so $\sigma_{Xe} > 0$. OLS is biased upward: it overstates the effect of education on health.
3. (a) $\mathbb{E}[ZY] = \beta \mathbb{E}[ZX] + \underbrace{\mathbb{E}[Ze]}_{=0}$, so $\beta = \text{Cov}(Z, Y) / \text{Cov}(Z, X)$.
 (b) $\hat{\beta}_{IV} = \frac{\sum Z_i Y_i}{\sum Z_i X_i} = \frac{\mathbf{Z}'\mathbf{Y}}{\mathbf{Z}'\mathbf{X}}$ (scalar case, demeaned).
4. (a) Row 1 = 2 × Row 2, so $\text{rank}(\mathbf{A}) = 1$. Not invertible ($\det = 4 - 4 = 0$).
 (b) No row is a multiple of the other, so $\text{rank}(\mathbf{B}) = 2$. Invertible ($\det = 6 - 1 = 5 \neq 0$).